

# Handwriting Author Identification with Convolutional Neural Nets

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## Background

- Forensic handwriting analysis is essential for identity & fraud detection
- Over 30% of USPIS cases involve analyses
- Can computer vision models make this faster & more objective?

**Goal:** Investigating the feasibility of predicting author identity and demographic information based on handwriting analysis using convolutional neural networks (CNNs).

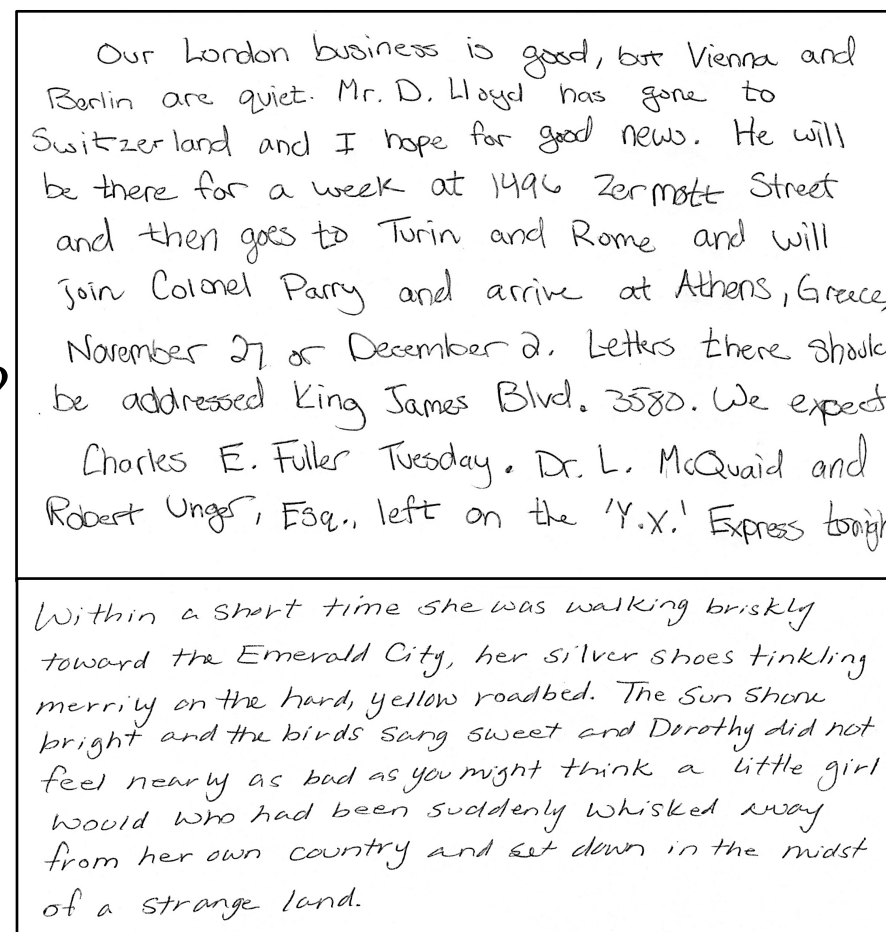


Figure 1: Example data images.

## Methods

- Transfer learning models with TensorFlow & Keras
- Evaluation with SciKit-Learn
- Computational GPU resources from Google Colab
- Feasibility models (MobileNetV2) hyperparameters:
  - 70 epochs
  - Batch size = 64
  - Image size 256x256x3

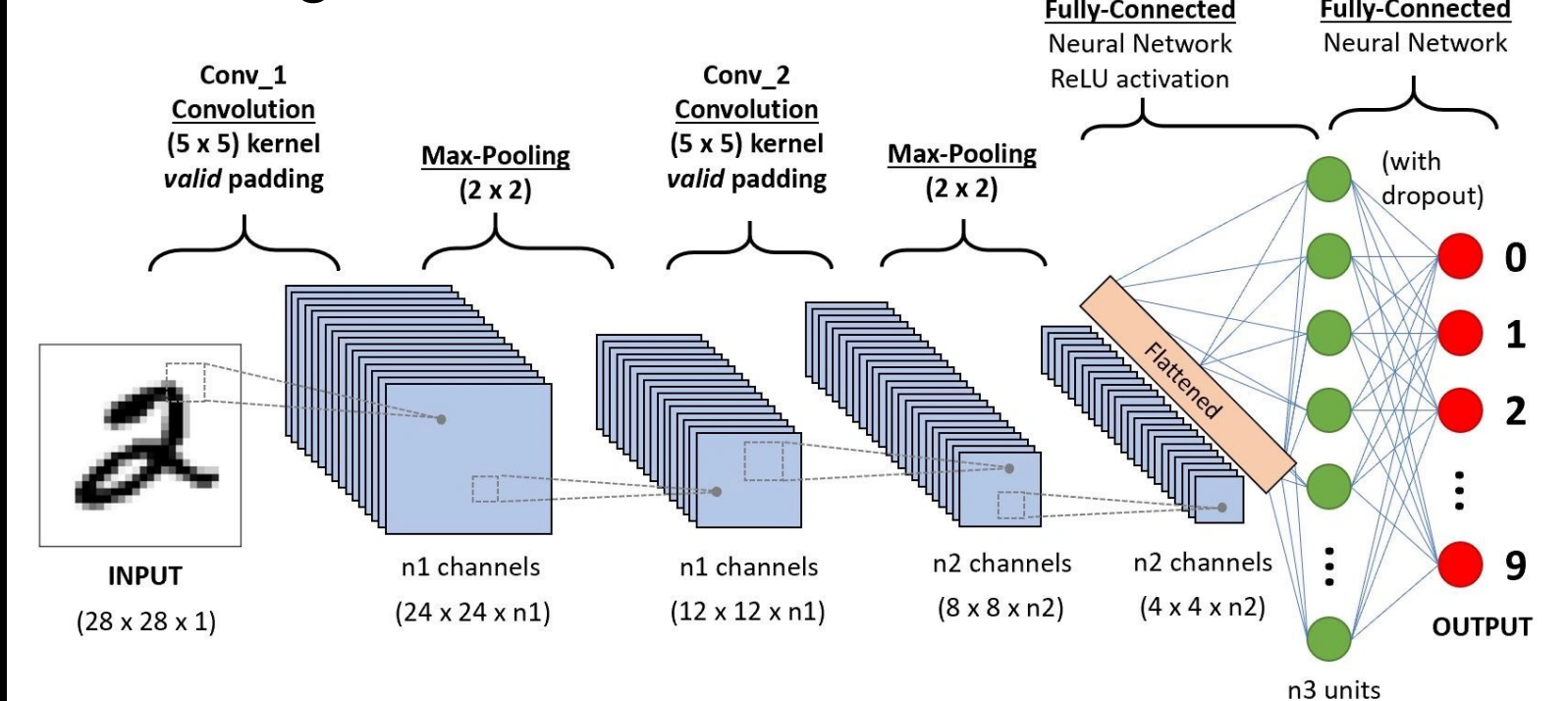


Figure 2: Convolutional neural net classification algorithm (Salar).

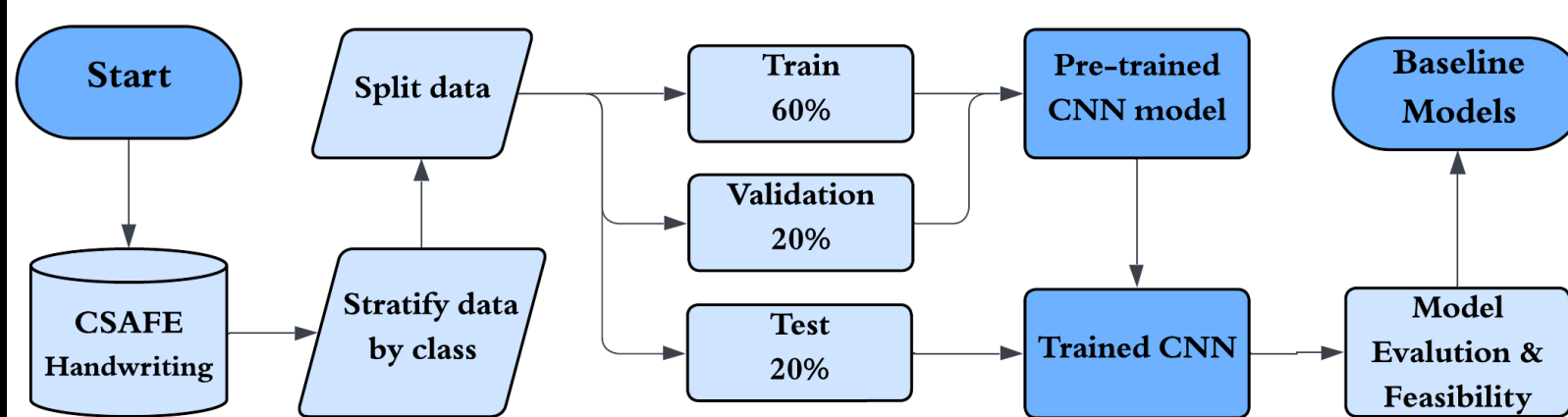


Figure 3: Baseline model pipeline flowchart.

## Results

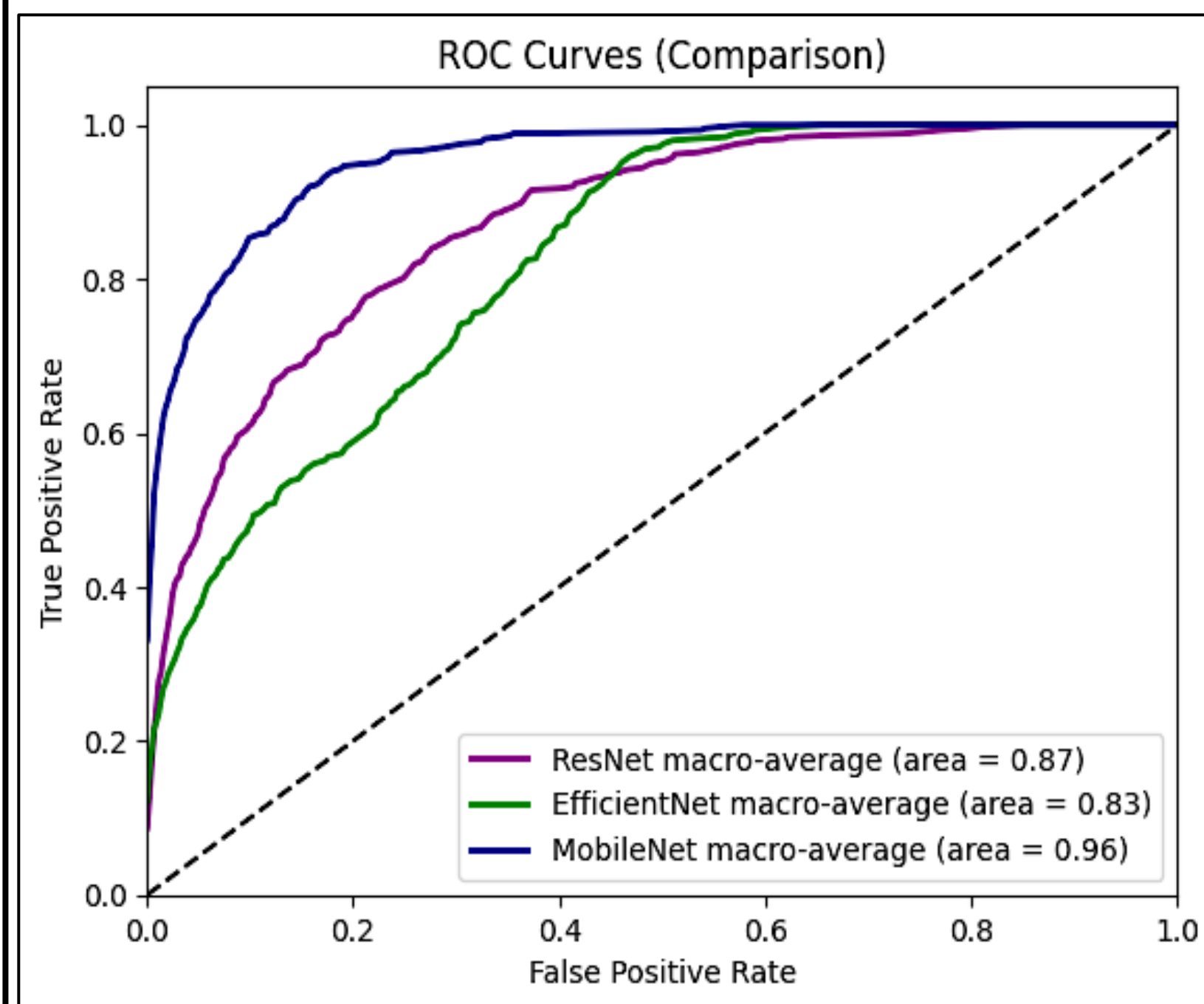


Figure 4: ROC-AUC plot comparing performance between three different preliminary CNN models.

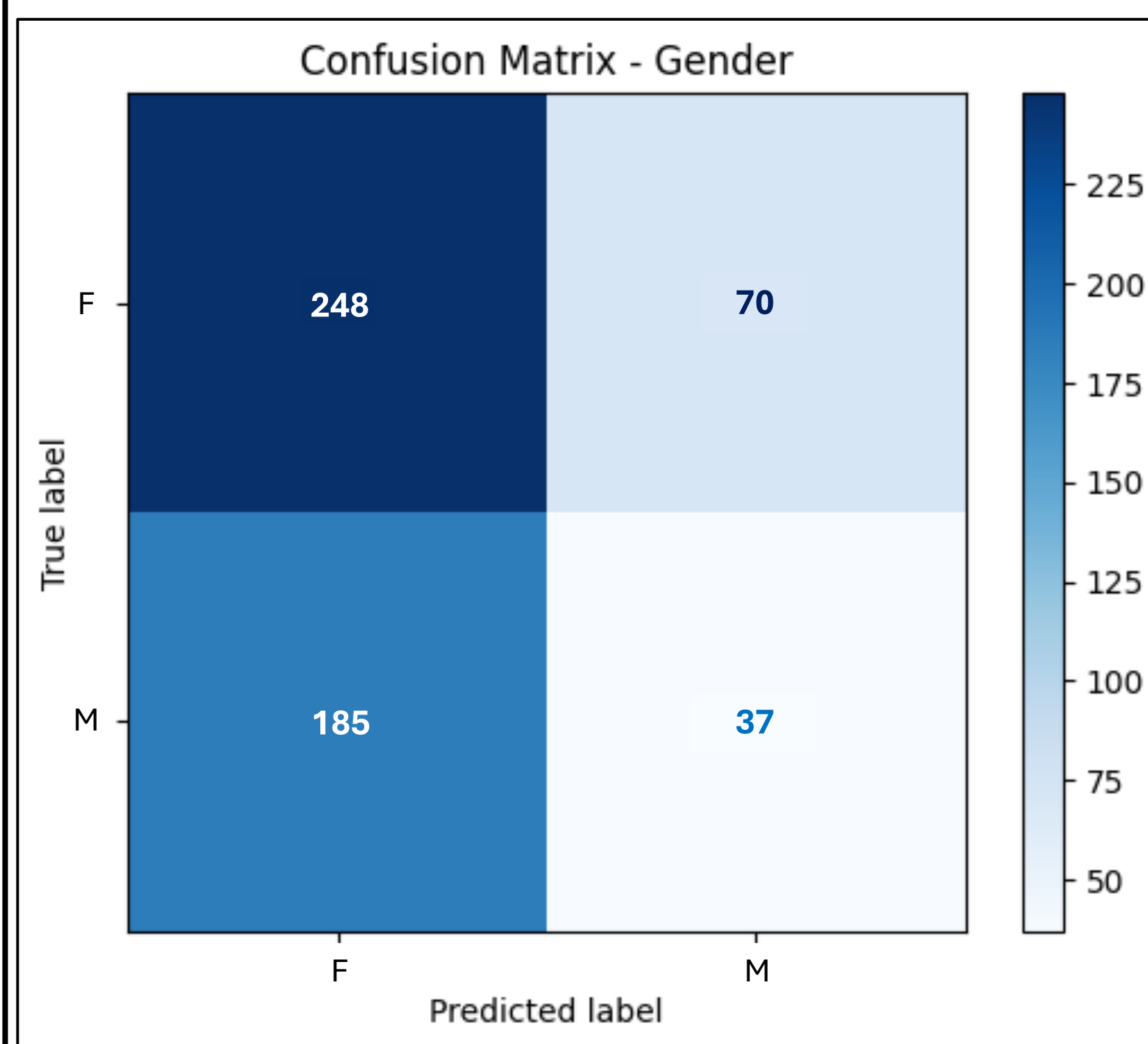


Figure 6: Confusion matrix of classification results on test set during gender prediction attempt.

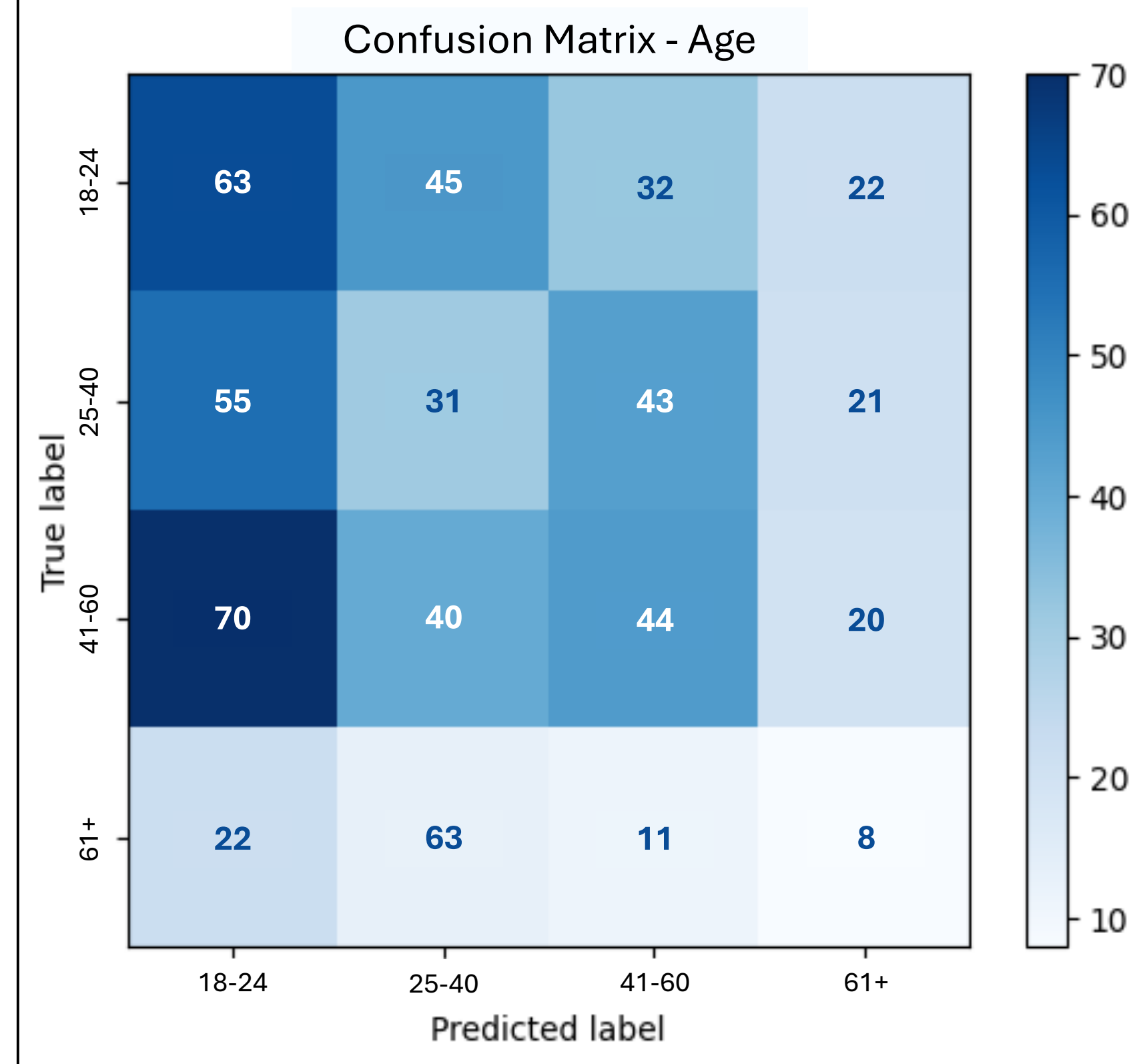


Figure 5: Confusion matrix of classification results on test set during age prediction attempt.

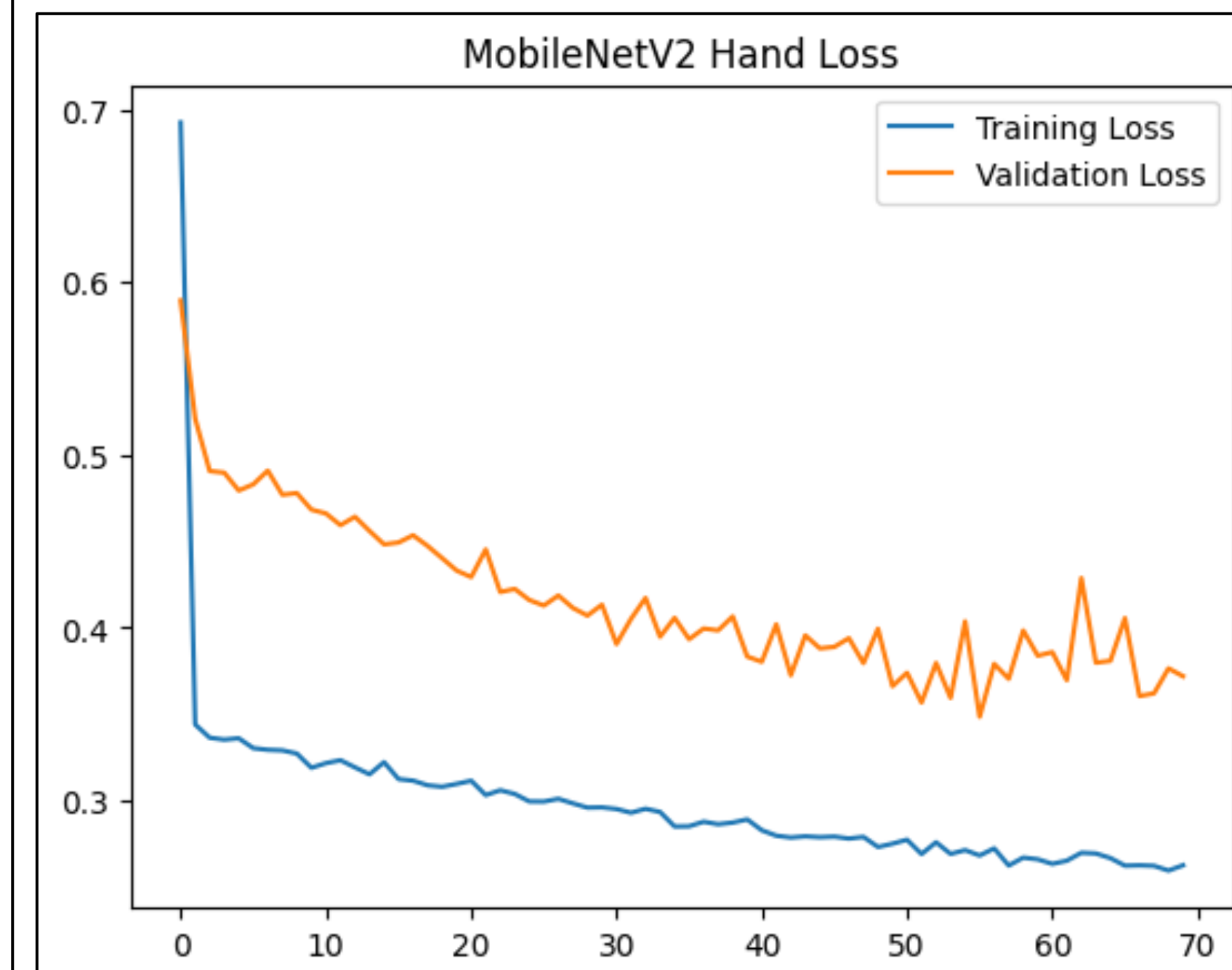


Figure 7: MobileNetV2 loss graph during training to predict handedness showing steep drop, indicating “memorization”.

## Discussion

- Demographic attribute prediction, such as for *handedness* and *age*, does not seem to be feasible based on these data
  - Models perform no better than random guessing
  - E.g., *handedness* prediction accuracy is approximately 90%, the same proportion as the majority class
  - Indicates overfitting, or label “memorization”
- Handwriting features *do* appear to be a viable predictor of *author identity*
  - Minor overfitting present (i.e., train acc. > test acc.)

Table 1: Metric comparison across target variables.

| Target Var. | Train Acc. | Test Acc. | Major Class % |
|-------------|------------|-----------|---------------|
| Gender      | 69.3%      | 72.2%     | 60%           |
| Age         | 64.7%      | 63.7%     | 33%           |
| Handedness  | 89.6%      | 89.1%     | 90%           |
| Author      | 86.0%      | 79.1%     | 1.1%          |

## Future work

- Handling class imbalance to re-evaluate feasibility
- Building custom CNNs
- Hyperparameter & model tuning
- Image augmentation
- Investigate prompt bias/data leakage
- Dimensionality reduction/projection

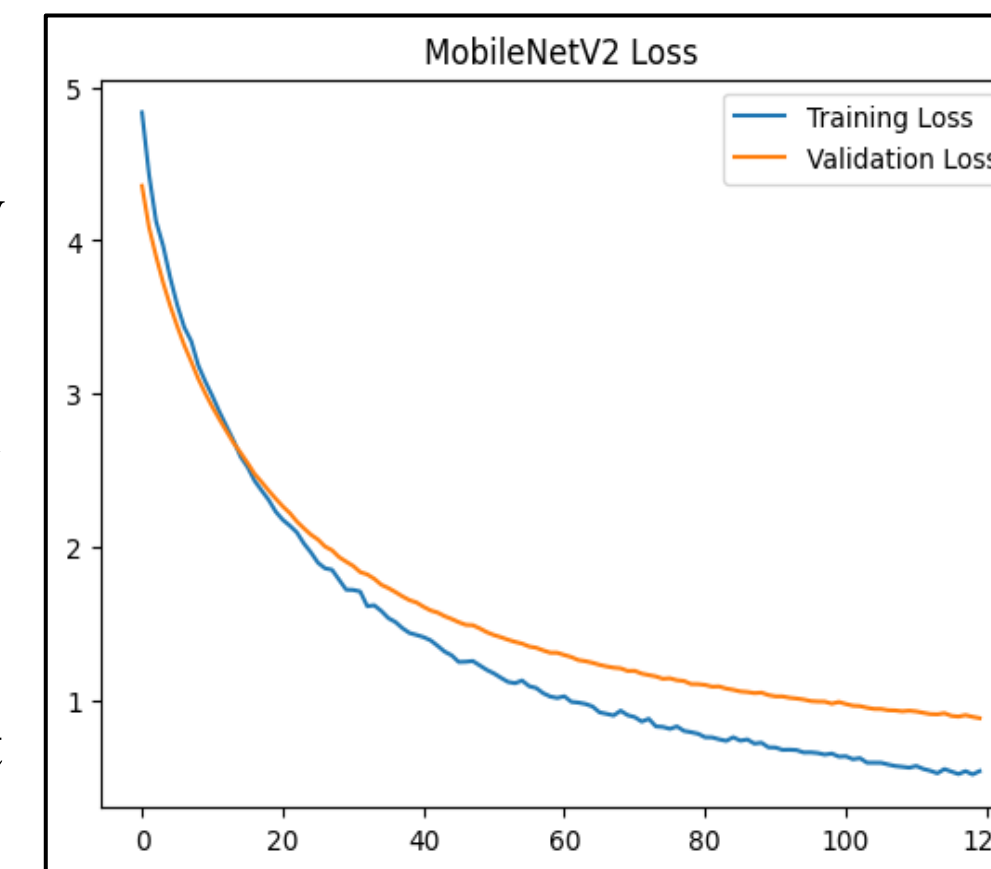


Figure 8: MobileNetV2 loss curve (most recent model attempt), predicting authorship.

## Citations

Crawford, A., Ray, A., Carriquiry, A., Kruse, J., & Peterson, M. (2019). CSAFE Handwriting Database. <https://doi.org/10.25380/u002Fiastate.10062203.v1>. Dataset.

Salar. “Quick Introduction to Convolutional Neural Networks (CNN).” Medium, Artificial Intelligence in Plain English, 1 Mar. 2023, [ai.plainenglish.io/quick-introduction-to-convolutional-neural-networks-cnn-ad4a6b0e1cb8](https://ai.plainenglish.io/quick-introduction-to-convolutional-neural-networks-cnn-ad4a6b0e1cb8). Image.